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P276

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[6003] 355

T.E. (Computer Engg.)

WEB TECHNOLOGY

(2019 Pattern) (Semester-II) (310252)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What are strengths of XML technology? Explain the need of XML. [5]
b) What are DTD's? Explain how do they work? [5]
c) Explain URL writing and cookies in servlet with example. [8]

OR

- Q2)** a) Explain servlet architecture in detail. [8]
b) What do you understand by AJAX? Explain it. [5]
c) Write a servlet which accepts two numbers using POST method and display the maximum number. [5]

- Q3)** a) List & describe important interceptors provided by strut framework. [5]
b) Identify & justify the benefits of using Web Services. [5]
c) Explain JSP life cycle with diagram. [7]

OR

- Q4)** a) What is JSP? Enlist advantages of JSP over servlet? [5]
b) What is WSDL and SOAP? Explain WSDL in detail. [5]
c) Draw and explain MVC architecture for developing web application. [7]

- Q5)** a) What is WAP? Explain components of WAP architecture in detail. [8]
b) What is multidimensional arrays in PHP? Explain it with simple PHP code. [5]
c) Explain overview of node JS. [5]

OR

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- Q6)** a) Explain how cookies and session are used for session management in PHP. [8]
b) What is WML? Explain WML elements. [5]
c) Explain in brief overview of ASP. NET. [5]

- Q7)** a) Explain how multiple selection constructs are implemented in Ruby. [7]
b) Explain Rails with AJAX in detail. [5]
c) Draw & explain the role of EJB container in Enterprise applications. [5]

OR

- Q8)** a) Explain how to write the methods and call the method in RUBY with example. [7]
b) What are the difference between java beans and EJB? [5]
c) What is Ruby programming language? List some features of Ruby. [5]



Solution:

Q1 a) What are strengths of XML technology? Explain the need of XML. (5 Marks)

Ans:

XML is a flexible, structured, platform-independent format used to store and exchange data between different systems.

Strengths of XML Technology (Any 4–5 points)

1. Platform & Language Independent

XML can be used on any operating system and with any programming language.

2. Self-descriptive Data Format

Tags explain the data themselves, so no external description is required.

3. Supports Hierarchical Data

Data is stored in a tree structure, making it easy to represent complex information.

4. Extensible Format

Users can create their own tags according to their application needs.

5. Separation of Data & Presentation

XML data can be displayed differently using XSLT, HTML, or CSS.

6. Widely Used in Web Services & APIs

SOAP, configuration files, Android manifests, etc., all use XML.

Need of XML (Any 3–4 points)

1. Standard Format for Data Exchange

Helps different systems share data in a universally understood format.

2. Data Validation

Using DTD or XML Schema ensures correctness and structure of data.

3. Interoperability Between Applications

Makes integration easy between apps, devices, and platforms.

4. Storage & Configuration

Many applications use XML for configuration (e.g., web.xml, Android XML files).

Q1 (b) What are DTDs? Explain how they work. [5 Marks]

Ans:

DTD (Document Type Definition) defines the rules, structure, and allowed elements/attributes of an XML document.

What are DTDs?

1. Definition of XML Structure

DTD specifies what elements and attributes can appear in the XML document.

2. Acts as a Blueprint

Works like a grammar for XML, ensuring a correct sequence of tags.

3. Two Types

- **Internal DTD** (written inside XML)
- **External DTD** (in separate .dtd file)

4. Used for Validation

Helps an XML parser confirm if the XML is valid or not.

How DTD Works (Step-by-Step)

1. XML Document is Linked to a DTD

Either internally or externally.

2. Parser Reads XML and DTD Together

The parser checks structure, tag names, and order.

3. Validation Process Happens

Each element in XML is compared with the rules in DTD.

4. Error is Shown if Structure Violates DTD

Ensures consistent and reliable XML data.

Q1 (c) Explain URL rewriting and cookies in servlet with example. [8 Marks]

Ans:

Session tracking in servlets helps identify users across multiple requests. Two common techniques: **Cookies** and **URL Rewriting**.

1. Cookies in Servlet

Meaning:

Cookies are small text files stored in the client's browser to maintain user identity.

How Cookies Work :

1. Server Creates a Cookie Object

Cookie c = new Cookie("username", "Vijay");

2. Cookie Sent to Browser

Browser saves the cookie.

3. Browser Sends Cookie Back Automatically

On every next request, cookie is returned.

4. Server Identifies User Using Cookie Data

Helps maintain user session.

Example (Servlet Code):

```
Cookie c = new Cookie("user","vijay");  
response.addCookie(c); // sending cookie
```

2. URL Rewriting in Servlet

Meaning:

URL rewriting adds session information directly into the URL when cookies are disabled.

How URL Rewriting Works (Points):

1. Session ID is added to URL manually

e.g.,

http://example.com/home?sessionid=12345

2. Client clicks the link containing session data

3. Server extracts session ID from URL

4. User is identified without using cookies

Example:

```
String url = response.encodeURL("home.jsp");  
out.println("<a href=\"" + url + "\">Go to Home</a>");
```

OR

Q2 (a) Explain servlet architecture in detail. [8 Marks]

Ans:

Servlet architecture defines how a servlet interacts with the web container to process client requests and send back responses.

Servlet Architecture

1. Client (Browser)

- User sends a request through a browser (HTTP request).
- Example: clicking a form or a link.

2. Web Server

- Receives the client request.
- Passes the request to the **Servlet Container**.

3. Servlet Container / Web Container

- Core component that manages servlet lifecycle.
- Examples: **Tomcat, Jetty, GlassFish**.

4. Responsibilities of Servlet Container:

- **Loading servlet class** into memory.
- **Creating servlet object** (only once).
- **Calling lifecycle methods:**
 - `init()` → initialization
 - `service()` → for request processing
 - `destroy()` → before shutting down
- Managing **multithreading**.
- Managing **session** (cookies, URL rewriting).
- Handling **security** and **mapping**.

5. Request Processing Flow:

- Container creates **HttpServletRequest** object.
- Container creates **HttpServletResponse** object.
- Calls `service()` which decides:
 - `doGet()` for GET request
 - `doPost()` for POST request

6. Servlet Generates Response:

- Servlet prepares HTML/text output.
- Response is sent to browser using `OutputStream` or `PrintWriter`.

7. Client Receives Output

- Browser displays the page generated by servlet.

Q2 (b) What do you understand by AJAX? Explain it. [5 Marks]

Ans:

AJAX allows web pages to update data **without refreshing the entire page**.

Explanation of AJAX

1. **AJAX = Asynchronous JavaScript and XML**

It is not a language; it is a technique using JavaScript + XML/JSON.

2. **Works Asynchronously**

Browser communicates with the server in the background while the page remains visible.

3. **No Page Refresh Required**

Only a portion of the webpage updates (similar to how Google search suggestions work).

4. **Uses XMLHttpRequest Object**

AJAX sends and receives data using this object (or Fetch API).

5. **Data Formats Supported**

XML, JSON, HTML, text.

6. **Better User Experience**

- Faster response
- Smooth UI
- Less server load

How AJAX Works

1. Browser generates an event (button click, typing).
2. JavaScript creates an **XMLHttpRequest** object.
3. Request sent to server asynchronously.
4. Server processes the request.
5. Server sends data (XML/JSON) back to browser.
6. JavaScript updates only the required part of the page.

Q2 (c) Write a servlet which accepts two numbers using POST method and displays the maximum number. [5 Marks]

Ans:

POST method is used when form data is sent securely and not shown in the URL.

Servlet Code (Perfect 5 Marks Answer)

HTML Form (to send data using POST):

```
<form action="MaxServlet" method="post">
    Number 1: <input type="text" name="n1"><br>
    Number 2: <input type="text" name="n2"><br>
    <input type="submit" value="Find Max">
</form>
```

Servlet Code:

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class MaxServlet extends HttpServlet {
    public void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        // Reading values
        int a = Integer.parseInt(request.getParameter("n1"));
        int b = Integer.parseInt(request.getParameter("n2"));

        int max = (a > b) ? a : b;

        out.println("<h2>Maximum Number is: " + max + "</h2>");
    }
}
```

@TheVijayAcademy

Q3) (a) List & describe important interceptors provided by Struts framework.[5]

Ans:

Important Struts2 Interceptors

1. Validation Interceptor

- Performs form/field validations before action executes.
- If validation fails → returns error page.

2. File Upload Interceptor

- Handles file upload logic.
- Validates file type, size, and stores uploaded file.

3. ServletConfig Interceptor

- Provides access to servlet API objects → request, response, session.
- Useful when you need low-level HTTP access.

4. Parameters Interceptor

- Maps incoming HTTP request parameters to Action class properties.
- Example: name, email → automatically stored in action fields.

5. Exception Interceptor

- Catches unhandled exceptions in action.
- Redirects user to error page without breaking application.

Q3) (b) Identify & justify the benefits of using Web Services.[5]

Ans:

Benefits of Web Services (point-to-point)

1. Platform Independent

- Java, .NET, Python — all can communicate via XML/SOAP/JSON.

2. Language Independent

- Data exchanged in XML/JSON → readable by any programming language.

3. Interoperability

- Applications running on different OS, networks can communicate easily.

4. Reusability

- Service can be reused across multiple applications (mobile, web, desktop).

5. Loose Coupling

- Client and server do not depend on internal implementation of each other.

Q3) (c) Explain JSP life cycle with diagram.[5]

Ans:

JSP lifecycle = steps followed from creation of JSP page → until destruction by container.

JSP Life Cycle

1. Translation Phase

- JSP file → translated into a Servlet Java file by container.

2. Compilation Phase

- Generated servlet .java file → compiled into .class file.

3. Class Loading

- Servlet class loaded into JVM.

4. Instantiation

- JSP servlet object is created.

5. Initialization (jspInit())

- Called once in lifetime.
- Used for resource initializations.

6. Request Processing (_jspService())

- Executes for every client request.
- Contains all code for generating dynamic response.

7. Destruction (jspDestroy())

- Called once before removing JSP instance.
- Used to release resources.

OR-Q4) (a) What is JSP? Enlist advantages of JSP over Servlet?[7]

Ans:

What is JSP?

- JSP (Java Server Pages) is a server-side technology used to create dynamic web pages using HTML + Java.

Advantages of JSP over Servlets (any 4–6 points)

1. Easier to write

- JSP uses HTML-like tags → simpler than full Java servlet code.

2. Separation of UI & Logic

- HTML in JSP, business logic in JavaBeans → clean structure.

3. Automatic Servlet Conversion

- JSP automatically becomes a servlet → no manual coding.

4. Implicit Objects

- request, response, session etc. available by default in JSP.

5. Faster changes

- Minor UI changes in JSP do not require recompiling full servlet.

6. Better suited for view layer

- Ideal for front-end presentation logic.

Q4) (b) What is WSDL and SOAP? Explain WSDL in detail.[5]

Ans:

SOAP

SOAP = Simple Object Access Protocol

- XML-based protocol used for sending structured messages between applications over HTTP.

WSDL

- WSDL = Web Services Description Language
- XML document that describes what a web service does and how to access it.

WSDL in Detail

1. Types Section

- Defines data types used by service (XML Schema).

2. Message Section

- Defines input and output messages (request/response structure).

3. PortType (Operations)

- Lists functions/methods exposed by web service.

4. Bindings

- Defines communication protocol (SOAP/HTTP).

5. Service Section

- Actual endpoint (URL) where service is hosted.

Q4) (c) Draw and explain MVC architecture for developing web application.[7]

Ans:

MVC = Model–View–Controller

Architecture used to separate business logic, UI, and request handling.

MVC – Explanation

1. Model

- Contains data, business logic, database operations.

2. View

- JSP/HTML pages displayed to user.

3. Controller

- Receives request, calls model, selects appropriate view.

Flow

1. User sends request to **Controller**.
2. Controller interacts with **Model** to process data.
3. Controller selects appropriate **View**.
4. View sends response to user.

Q5 a) What is WAP? Explain components of WAP architecture in detail. (8 marks)

Ans:

WAP (Wireless Application Protocol) is a standard that allows mobile phones to access internet services like email, browsing, online apps. It was used in old mobile devices before smartphones became common.

WAP Architecture Components

1) Mobile Device

- The user's mobile phone or handheld device.
- Runs a **WAP micro-browser** that displays WML pages.
- Sends requests using wireless networks.

2) WAP Micro-browser

- A lightweight browser specially made for low power devices.
- Works with **limited screen size, low bandwidth, and small memory**.
- Reads and displays **WML pages** instead of HTML.

3) Wireless Network

- Used to transmit mobile requests.
- Can be GSM, CDMA, GPRS, 3G etc.
- Transfers data between mobile and WAP gateway.

4) WAP Gateway / Proxy

This is the most important component.

Functions:

- Converts **WAP request** → **HTTP request**
- Converts **HTTP response** → **WAP response**
- Encodes and compresses the data to reduce size
- Handles security and authentication

It acts as a bridge between mobile device and the internet.

5) Internet / Web Server

- Stores website content, WML pages, PHP/ASP scripts etc.
- Receives HTTP request (from gateway) and sends back response.

6) Application Server

- Runs the business logic (PHP, ASP.NET, Java).
- Generates dynamic content.
- Sends output to web server → gateway → mobile device.

Flow of WAP Communication

1. Mobile sends request using WAP micro-browser
2. WAP Gateway converts to HTTP
3. Web server processes the request
4. Response goes back to gateway
5. Gateway converts to WML
6. Mobile device displays result

Q5 b) What is multidimensional array in PHP? Explain with simple PHP code. (5 marks)

Ans:

A multidimensional array stores **multiple arrays inside a single array**.
Useful for representing tables, records, student lists etc.

Explanation

1) Definition

A multidimensional array is an array containing one or more arrays.

2) 2D Array (array of arrays)

Useful for storing rows and columns (like a table).

3) Syntax

```
$array = array(  
    array(value1, value2),  
    array(value3, value4)  
);
```

Example Code

```
$students = array(  
    array("101", "Amit", "CSE"),  
    array("102", "Riya", "IT"),  
    array("103", "Karan", "ENTC")  
);  
  
echo $students[0][1]; // Output: Amit
```

4) Purpose

- Represent database-like tables
- Store structured data
- Easy for looping through nested values

Q5 c) Explain overview of Node.js. (5 marks)

Ans:

Node.js is a **server-side JavaScript runtime** that allows JavaScript to run outside the browser.

Overview

1) Based on V8 Engine

- Uses Google Chrome's **V8 JavaScript Engine**.
- Very fast execution.

2) Event-driven & Non-blocking I/O

- Handles many requests at the same time.
- No waiting for operations like file read or database fetch.
- Ideal for real-time apps.

3) Uses JavaScript on Server

- Same language for frontend and backend.
- Developers can write full-stack apps using JS only.

4) Lightweight and Scalable

- Suitable for chat apps, real-time apps, API servers.

5) Rich Package Ecosystem (npm)

- Largest library ecosystem.
- Millions of reusable modules.

6) Used for

- APIs
- Web apps
- Streaming apps
- Real-time chat
- IoT applications

OR - Q6 a) Explain how cookies and sessions are used for session management in PHP. (8 marks)

Ans:

Session management keeps track of user data across multiple pages (example: login, cart).

PHP uses **Cookies** and **Sessions** to manage it.

1) Cookies in PHP

Definition

A cookie is a small piece of data stored in the **user's browser**.

Use

- Store non-sensitive information
- Remember last login name, theme, preferences

Setting a Cookie

```
setcookie("user", "Vijay", time()+3600);
```

Accessing a Cookie

```
echo $_COOKIE['user'];
```

2) Sessions in PHP

Definition

A session stores information on the **server side** for each user.

Use

- Login systems
- Shopping carts
- Secure transactions

Starting a Session

```
session_start();
```

Storing Data

```
$_SESSION['username'] = "Vijay";
```

Reading Data

```
echo $_SESSION['username'];
```

Destroying Session

```
session_destroy();
```

Q6 b) What is WML? Explain WML elements. (5 marks)

Ans:

WML (Wireless Markup Language) is a markup language used to create WAP pages for early mobile phones.

WML Elements

1) <wml>

- Root element
- Contains all cards

2) <card>

- Represents a single page (screen)
- Has attributes like id, title

3) <p>

- Paragraph element
- Used for text display

4) <anchor>

- Creates a hyperlink
- Similar to <a> in HTML

5) <input>

- Takes user input
- Example: text box, password

6) <do>

- Defines actions (go, prev, accept)
- Similar to buttons

Q6 c) Explain in brief overview of ASP.NET. (5 marks)

Ans:

ASP.NET is a **Microsoft web development framework** used to build dynamic websites and web applications.

Overview

1) Part of .NET Framework

- Uses CLR (Common Language Runtime)
- Supports multiple languages: **C#, VB.NET**

2) Server-side Technology

- Code runs on the server
- Output (HTML) sent to browser

3) Rich Controls

- TextBox
- Button
- GridView
- DropDownList

4) Built-in Features

- Security
- Authentication
- Session management
- State management

5) Development Models

- Web Forms
- MVC
- Web API
- Razor Pages

6) Used for

- Websites
- Enterprise apps
- Web services
- APIs

Q7 (a) Explain how multiple selection constructs are implemented in Ruby. [7]

Ans:

Multiple-selection constructs allow Ruby to choose **one option from many**. Ruby provides:

1. if–elsif–else

Used when multiple conditions must be checked one by one.

Syntax

```
if condition1
  # code
elsif condition2
  # code
elsif condition3
  # code
else
  # default
end
```

Example

```
marks = 75
```

```
if marks >= 90
  puts "Grade A"
elsif marks >= 75
  puts "Grade B"
elsif marks >= 60
  puts "Grade C"
else
  puts "Fail"
end
```

2. case-when

Cleaner and shorter than if-elsif chain.

Used when comparing one expression with many values.

Syntax

```
case variable
when value1
  # code
when value2
  # code
else
  # default
end
```

Example

```
day = 3
```

```
case day
when 1
  puts "Monday"
when 2
  puts "Tuesday"
when 3
  puts "Wednesday"
else
  puts "Other day"
end
```

3. Using ranges in case

score = 82

```
case score
when 90..100 then puts "Excellent"
when 75..89  then puts "Good"
when 50..74  then puts "Average"
else        puts "Fail"
end
```

Conclusion

Ruby supports multiple selection using:

- **if–elsif–else**
- **case–when**
- **ranges inside case**

These make decision-making simple and readable.

Q7 (b) Explain Rails with AJAX in detail. [5]

Ans:

AJAX (Asynchronous JavaScript and XML) allows a web page to update **without reloading** the whole page.

Rails supports AJAX using:

- Unobtrusive JavaScript (UJS)
- remote: true
- Rails responders

How Rails works with AJAX

1. AJAX Request

A form or link with `remote: true` sends the request asynchronously.

Example:

```
<%= form_with url: "/students", remote: true do |f| %>
  <%= f.text_field :name %>
  <%= f.submit "Save" %>
<% end %>
```

This form submits using AJAX, not a full page reload.

2. Controller Responds with JS

Controller returns JavaScript file (.js.erb).

```
def create
  @student = Student.create(student_params)
  respond_to do |format|
    format.js
  end
end
```

3. JS File Updates the Page Dynamically

File: create.js.erb

```
alert("Student added successfully!");
```

Or insert new HTML:

```
$("#student_list").append("<%= j render @student %>");
```

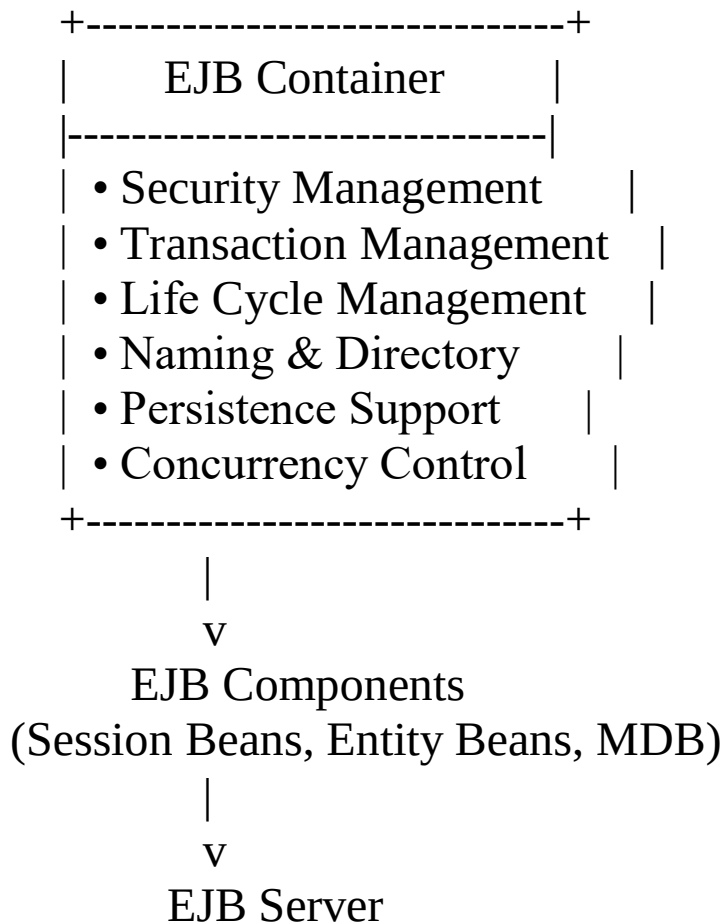
Advantages of Rails with AJAX

- ✓ Faster page updates
- ✓ No full page refresh
- ✓ Better user experience
- ✓ Clean separation using `.js.erb` files

Q7 (c) Draw & explain the role of EJB container in Enterprise applications. [5]

Ans:

Diagram



Role of EJB Container

The EJB container sits between the **EJB components** and the **application server**. It provides system-level services automatically:

1. Security Management

Controls authentication & authorization.

2. Transaction Management

Ensures data consistency using COMMIT or ROLLBACK.

3. Life Cycle Management

Creates, pools, and destroys EJB objects.

4. Naming & Directory Services

Uses JNDI to locate beans.

5. Persistence

Helps in storing data in database.

6. Concurrency Control

Allows many users to access application safely.

7. Remote Access

Supports distributed applications.

OR-Q8 (a) Explain how to write methods and call methods in Ruby with example. [7]

Ans :

Defining a Method

Use the def keyword.

```
def greet  
  puts "Hello Ruby!"  
end
```

Calling a Method

```
greet
```

Method with Parameters

```
def add(a, b)  
  return a + b  
end
```

```
result = add(5, 3)  
puts result
```

Method with Default Parameter

```
def welcome(name="Guest")  
  puts "Welcome #{name}"  
end
```

```
welcome("Ravi")  
welcome
```

Method Returning Multiple Values

```
def calc(a, b)  
  return a+b, a*b  
end
```

```
sum, product = calc(3, 4)  
puts sum  
puts product
```

Q8 (b) Difference between Java Beans and EJB. [5]

Ans:

Java Beans	Enterprise Java Beans (EJB)
Used for client-side UI components	Used for server-side enterprise logic
Runs inside normal JVM	Runs inside EJB Container
No built-in security or transactions	Provides security & transaction management
Simple, lightweight	Heavy-weight, enterprise level
Not distributed	Supports distributed applications

Q8 (c) What is Ruby programming language? List some features of Ruby. [5]

Ans:

What is Ruby?

Ruby is a **pure object-oriented, dynamic, general-purpose scripting language** created by Yukihiro Matsumoto.

It is simple, readable, and productive.

Features of Ruby

1. **Pure Object-Oriented** – Everything is an object.
2. **Flexible & Dynamic** – Variables and methods can be changed at runtime.
3. **Simple & Readable Syntax** – Easy to learn and write.
4. **Automatic Memory Management** – Garbage collection included.
5. **Rich Standard Library** – Many built-in classes and modules.
6. **Supports Modules & Mixins** – Multiple inheritance through mixins.
7. **Platform Independent** – Runs on Windows, Linux, macOS.
8. **Used in Rails** – Very powerful for web development using Ruby on Rails.

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